Christopher Collazo, M.S.
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https://nervecenter.github.io
Work Status: US citizen
Location: St. Petersburg, FL

Work Experience

Data & Software Engineer — St. Petersburg, FL
GeoSonics, Inc. — *November 2022 – Present*

- Lead engineer for a high-performance seismograph analysis engine:
  - Performs backend compute for industry-leading web-based vibration analysis data visualizations.
  - Native- and statically-compiled with Nim, musl-libc, and gnuplot.
  - Implements a number of hand-tuned vibration and signals analysis algorithms to a legally defensible standard.
  - Renders professional reports using Gnuplot for SVG data visualization and HTML/CSS for layout.
  - Investigated and reverse-engineered obfuscated C code to verify and reimplement legacy computations.
  - Utilized Nim's innovative value semantics for high-performance, memory safety, and structural simplicity.
  - Documented using modern typesetting tools such as Typst and Markdown.
- Also lead engineer for:
  - GUI application for RS232 serial hardware configuration using Godot and GDScript.
  - Prototyping next-generation record data containers utilizing MessagePack.

IT Software Engineer — Tampa, FL
University of South Florida — *January 2017 – November 2022*

- Lead engineer for a full-stack asset management database application using Clojure & Datomic backend.
- Administered and documented for the USF GAIVI GPU HPC cluster.
- Graduate assistant: Lecturing, grading, tutoring, and administering curricula for graduate-level courses.

Co-Founder, Principal Software Engineer — Tampa, FL
Sigil Technologies, Inc. — *October 2015 – August 2016*

- Lead engineer, business management, and sales for a SaaS full-stack application built with Clojure, PostgreSQL, JavaScript, and jQuery.

Education

M.S. in Computer Science, focus AI & Machine Learning — Tampa, FL
University of South Florida — *May 2021*

- **Alfred P. Sloan Foundation Minority Ph.D. Scholar**

B.S. in Computer Science — Tampa, FL
University of South Florida — *May 2015*

- **Dean's List** · USF College of Engineering · Fall 2014 & Spring 2015

Open Source & Presentations

The Virtues of Values (NimConf 2026) - https://www.youtube.com/watch?v=Ki4-kbSCnXs

- Describing how Nim's value and functional semantics enable clean, safe, and performant program structure.

nimgnuplot - https://github.com/nervecenter/nimgnuplot

- A Nim library for generating gnuplot plots from Datamancer dataframes.

Selected Works

- **Collazo, C.**, Vargas, I., Cara, B., Weinheimer, C. J., Grabau, R. P., Goldgof, D., Hall, L., Wickline, S. A., & Pan, H. (2024). *Synergizing Deep Learning-Enabled Preprocessing and Human–AI Integration for Efficient Automatic Ground Truth Generation.* MDPI Bioengineering, 11(5), 434. https://doi.org/10.3390/bioengineering11050434
  - Summary: We formulate a "self-improvement" process for a convolutional segmentation model wherein it iteratively normalizes and retrains on its training data, called *bootstrapping*, improving its accuracy over time.
  - Source code: https://github.com/nervecenter/bsp-experiment
- **Collazo, C.** (2021). *Active Learning and Preprocessing for Medical Histopathology - A Literature Review.* University of South Florida. Major area paper & presentation.
- **Collazo, C.**, Vargas, I., Cara, B., Weinheimer, C. J., Grabau, R. P., Goldgof, D., Hall, L., Wickline, S. A., & Pan, H. (2019). *Transferrable Deep Learning Algorithms for Identifying Area At Risk in Acute Myocardial Infarction.* Abstract. Biomedical Imaging and Instrumentation, Deep Learning in Microstructural Imaging (Oral), BMES Annual Meeting 2019. https://submissions.mirasmart.com/BMESArchive/AdvancedSearch.aspx

Skills

Expert

- Building native- and statically-compiled CLI applications and libraries for server and embedded deployment.
  - C, Nim, Bash, Make, musl-libc, gcc toolchain, ARM & RISC-V cross-compilation
- Performing scientific analysis of large datasets.
  - Including statistical analysis, signal analysis, image/signal processing, deep learning, datavis, & dataframes.
  - Data such as: 1D vibration signals; environmental sensor data; 2D tissue sample images.
  - C, Python, Numpy, Pandas, PIL, Matplotlib, Seaborn, SciKit, Nim, Datamancer, gnuplot, Highcharts, OpenCV
- Document generation, typesetting, image processing, and graphic design.
  - LaTeX, Markdown, Typst, HTML, CSS, GIMP, RelaxedJS, Jinja/Nimja
- Building full-stack web and database applications.
  - Clojure, PostgreSQL, Datomic, JavaScript, TypeScript, jQuery, Zepto.js, HTML, CSS, Pico CSS, Bootstrap
- Formulating document specs and utilizing data container formats.
  - Text (JSON, CSV, XML, KDL), binary (MessagePack, UBJSON, proprietary)
- Version control.
  - Git, GitHub, GitLab, Bitbucket

Proficient

- Training and deploying convolutional deep learning models.
  - Python, PyTorch, Tensorflow, Keras, Numpy, SciKit, Numba, and Nvidia CUDA
- Professional and hobby experience building programs with the powerful Lisp family of languages.
  - Common Lisp, Clojure, Janet, Racket, Scheme
- Building cross-platform GUI applications with Godot, GDScript, and GDExtension.
  - Experience compiling minified slim Godot builds using C++ toolchains for small target distribution.

Intermediate

- Administering Linux HPC cluster systems and Python package deployments.
  - Bash, SSH, SGE, SLURM, Docker, Anaconda

Miscellaneous

- Past polyglot experience with numerous programming languages.
  - C#, C++, D, Elvish, F#, Haskell, Io, Java, Julia, OCaml, Rust, and more.